

Langfuse Implementation for BM25 & Hybrid Search

Added observability/monitoring to two search routes using Langfuse (a tracing service).

Route 1: BM25 Search (/api/search/bm25)

What happens when you search:

- ✓ Create a trace — Start recording the request
- ✓ Wrap the search — Capture what MongoDB is doing (the query pipeline)
- ✓ Record results — Log how many results came back
- ✓ Send to Langfuse — Upload the recording to the dashboard
- ✓ Send response to user — Return search results to the UI

☐	2026-07-31 18:54:55	↔	bm25-search	bm25-search	{{"\$search":{"index":"bm25_testcase_index","tex...	{{"_id":{"buffer":{"0":106,"1":5
☐	2026-07-31 18:54:54	↔	bm25-search	bm25-search	{"query":"Share Diagnostic Reports with Patients ...	{{"_id":{"buffer":{"0":106,"1":5

What The Dashboard Shows

When you make a search request, Langfuse records:

- Query text
- Search results (10 items, 5 items, etc.)
- Time taken
- MongoDB pipeline used

→ bm25-search :

+ Add to datasets

Annotate

▼

Add comment

2026-07-31 18:54:54.064

Env: default

Preview

Scores

Log View

Formatted

JSON

Input

Path	Value
query	"Share Diagnostic Reports with Patients via WhatsApp"
filters	empty object

Output

Path	Value
> 0	12 items
> 1	12 items
> 2	12 items
> 3	12 items
> 4	12 items
> 5	12 items
> 6	12 items
> 7	12 items
> 8	12 items

Route 2: Hybrid Search (/api/search/hybrid)

What happens when you search:

- ✓ Create a trace — Start recording
- ✓ Do BM25 search — Record keyword search step
- ✓ Generate embedding — Record Mistral AI embedding call (cost + tokens used)
- ✓ Do vector search — Record semantic search step
- ✓ Combine results — Record how results were merged (score fusion event)
- ✓ Send to Langfuse — Upload all the recordings
- ✓ Send response to user — Return combined results to the UI

☐	2026-07-31 19:15:35	🕒	score-fusion	hybrid-search	{"bm25Weight":0.5,"vectorWeight":0.5}	{"foundInBoth":5,"foundInBm2
☐	2026-07-31 19:15:34	↔	vector-search	hybrid-search	[{"\$vectorSearch":{"queryVector":[-0.016479492...	[{"_id":{"buffer":{"0":106,"1":5
☐	2026-07-31 19:15:33	🌀	generate-embedding	hybrid-search	Share Diagnostic Reports with Patients via Whats...	{"tokens":14,"cost":0.0000014
☐	2026-07-31 19:15:32	↔	bm25-search	hybrid-search	[{"\$search":{"index":"bm25_testcase_index","tex...	[{"_id":{"buffer":{"0":106,"1":5
☐	2026-07-31 19:15:29	↔	hybrid-search	hybrid-search	{"query":"Share Diagnostic Reports with Patients ...	[{"_id":{"buffer":{"0":106,"1":5

What The Dashboard Shows

- Query text
- BM25 results
- Mistral embedding cost & tokens
- Vector search results

- How results were fused together
- Final combined results
- Time for each step

generate-embedding

+ Add to datasets

Annotate

> Playground

Add comment

2026-07-31 19:15:33.004

Latency: 1.03s

Env: default

0 prompt → 0 completion (Σ 14)

mistral-embed

Preview

Scores

Log View

Formatted JSON

Input

Share Diagnostic Reports with Patients via WhatsApp

Output

Path	Value
tokens	14
cost	0.0000014000000000000001

Corrected Output

JSON

Real Example

Search: "Share Diagnostic Reports with Patients"

BM25 trace shows:

Query: "Share Diagnostic Reports with Patients"

Results: 10 test cases found

Time: 374ms

Status: ✔ Flushed to Langfuse

Hybrid trace shows:

Query: "Share Diagnostic Reports with Patients"

Weights: 50% BM25, 50% Vector

BM25 step: Found 10 cases (374ms)

Embedding: 14 tokens, cost \$0.0000014

Vector step: Found 15 cases (580ms)

Fused results: 12 final cases

- 5 found in BOTH (BM25 + Vector)
- 4 found in BM25 only
- 3 found in Vector only

Status:  Flushed to Langfuse

Bottom Line

-  We added visibility to what's happening inside BM25 and Hybrid searches
-  Nothing broke — searches work exactly the same as before
-  Langfuse dashboard now shows detailed audit trail of every search
-  Zero impact on test-case generator or any other feature